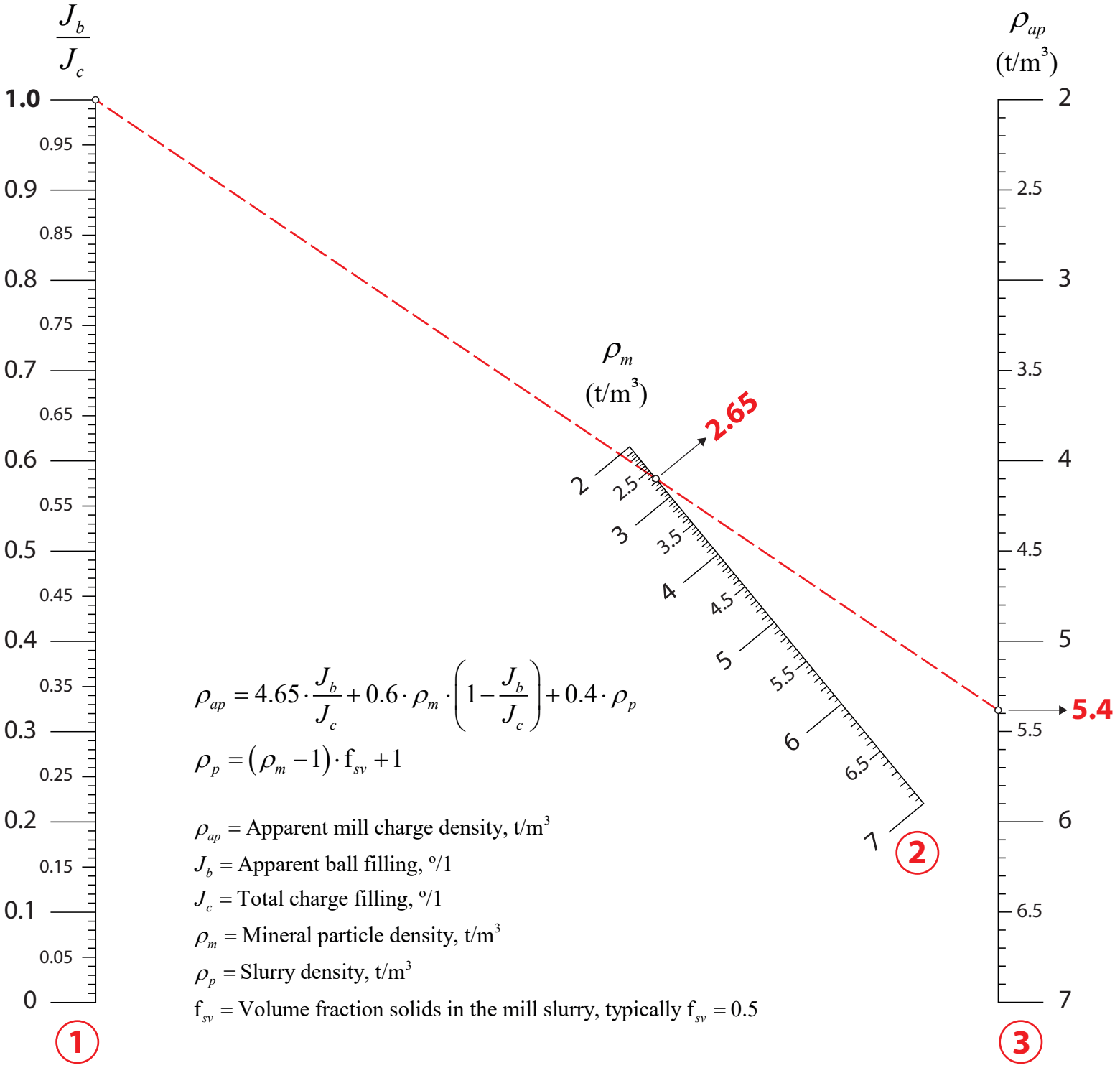
Apparent Mill Charge Density



$$\rho_{ap} = 4.65 \cdot \frac{J_b}{J_c} + 0.6 \cdot \rho_m \cdot \left(1 - \frac{J_b}{J_c}\right) + 0.4 \cdot \rho_p$$

$$\rho_p = (\rho_m - 1) \cdot f_{sv} + 1$$

ρ_{ap} = Apparent mill charge density, t/m^3

J_b = Apparent ball filling, %/1

J_c = Total charge filling, %/1

ρ_m = Mineral particle density, t/m^3

ρ_p = Slurry density, t/m^3

f_{sv} = Volume fraction solids in the mill slurry, typically $f_{sv} = 0.5$